Crank-slider mechanism

This schematic shows a crank pin following one full circular path and driving a piston through a fixed-length connecting rod. Its time parameterization is non-uniform; the equations use only existing AMX arithmetic, `abs`, conditional expressions, and `sqrt`, with no trigonometric functions.

```
let crankRadius: Number = 34 let rodLength: Number = 105 let pivotX: Number = 70 fn crankX(t: Number, r: Number): Number = r * (1 - 2 * abs(t - 1)) fn crankY(t: Number, r: Number): Number = if t <= 1 then sqrt(r * r - crankX(t, r) * crankX(t, r)) else -sqrt(r * r - crankX(t, r) * crankX(t, r)) fn sliderX(t: Number, pivot: Number, r: Number, rod: Number): Number = pivot + crankX(t, r) + sqrt(rod * rod - crankY(t, r) * crankY(t, r)) diagram crankSlider {   title: "Crank-slider mechanism"   description: "A crank pin follows one full circular path and drives a piston through a fixed-length connecting rod."   width: 320   height: 200   time t from 0 to 2 duration 4   line lowerGuide {     x1: 115     y1: 88     x2: 260     y2: 88     stroke: "#52616B"     lineWidth: 2     arrow: "none"   }   line upperGuide {     x1: 115     y1: 112     x2: 260     y2: 112     stroke: "#52616B"     lineWidth: 2     arrow: "none"   }   line connectingRod {     x1: pivotX + crankX(t, crankRadius)     y1: 100 + crankY(t, crankRadius)     x2: sliderX(t, pivotX, crankRadius, rodLength)     y2: 100     stroke: "#D07A32"     lineWidth: 5     arrow: "none"   }   line crank {     x1: pivotX     y1: 100     x2: pivotX + crankX(t, crankRadius)     y2: 100 + crankY(t, crankRadius)     stroke: "#6950A1"     lineWidth: 4     arrow: "none"   }   circle pivot {     cx: pivotX     cy: 100     r: 7     stroke: "#263238"     fill: "#FFFFFF"     lineWidth: 2   }   circle crankPin {     cx: pivotX + crankX(t, crankRadius)     cy: 100 + crankY(t, crankRadius)     r: 6     stroke: "#263238"     fill: "#F3B64B"     lineWidth: 2   }   rectangle piston {     x: sliderX(t, pivotX, crankRadius, rodLength) - 12     y: 90     width: 24     height: 20     stroke: "#263238"     fill: "#DCE8F0"     lineWidth: 2   }   text timeLabel {     x: 18     y: 24     value: "t = ${round(t, 2)} s"     fill: "#263238"   } }
```

Crank-slider mechanism

A crank pin follows one full circular path and drives a piston through a fixed-length connecting rod.



t = 0 s

The crank equations parameterize a complete revolution over `t = 0` to `t = 2`. The rod-length constraint determines the slider position directly for every selected time; the document does not model accumulated simulation state.